

5.3 Transformations of Sine and Cosine Worksheet #2

MCR3U

SOLUTIONS

1) A sinusoidal function has an amplitude of 5 units, a period of 120° , and a maximum at $(0, 3)$.

a) Represent the function with an equation using a sine function

$$a = 5$$

$$d_{\sin} = d_{\cos} - \frac{90}{k} = 0 - \frac{90}{3} = -30$$

$$k = \frac{360}{\text{period}} = \frac{360}{120} = 3$$

$$c = \text{max} - |a| = 3 - 5 = -2$$

$$y = 5 \sin[3(x+30)] - 2$$

b) Represent the function with an equation using a cosine function

$$d_{\cos} = 0$$

$$y = 5 \cos(3x) - 2$$

2) A sinusoidal function has an amplitude of $\frac{1}{2}$ units, a period of 720° , and a maximum at $(0, \frac{3}{2})$.

a) Represent the function with an equation using a sine function

$$a = 0.5$$

$$d_{\sin} = d_{\cos} - \frac{90}{|k|} = 0 - \frac{90}{0.5} = -180$$

$$k = \frac{360}{\text{period}} = \frac{360}{720} = 0.5$$

$$c = \text{max} - |a| = 1.5 - 0.5 = 1$$

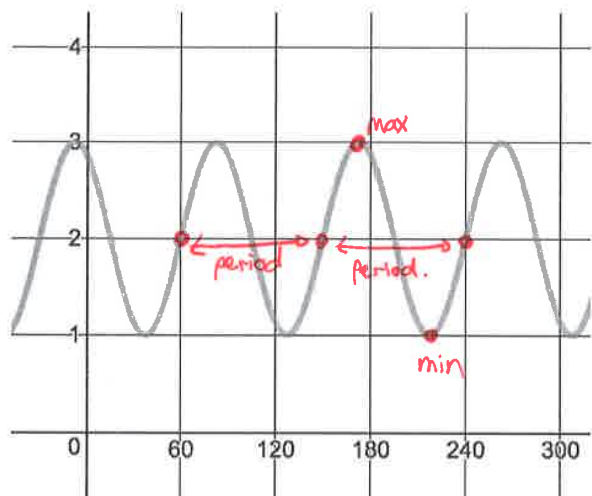
$$y = 0.5 \sin[0.5(x+180)] + 1$$

b) Represent the function with an equation using a cosine function

$$d_{\cos} = 0$$

$$y = 0.5 \cos[0.5x] + 1$$

3) Determine the equation of a cosine function that represents the graph shown.



$$a = \frac{\text{max} - \text{min}}{2} = \frac{3 - 1}{2} = 1$$

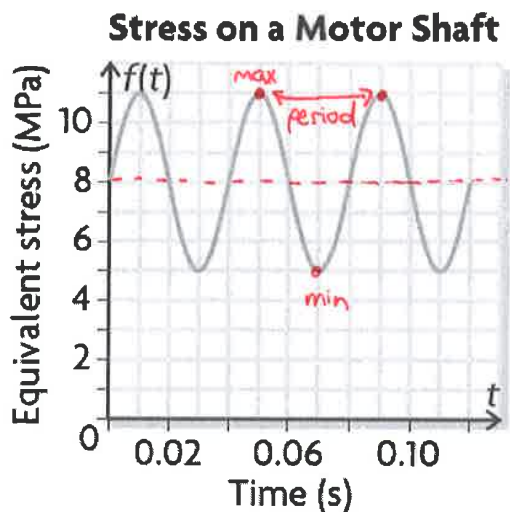
$$k = \frac{360}{\text{period}} = \frac{360}{150 - 60} = \frac{360}{90} = 4$$

$$c = \text{max} - |a| = 3 - 1 = 2$$

$$d_{\cos} = d_{\sin} + \frac{90}{k} = 60 + \frac{90}{4} = 82.5$$

$$y = \cos[4(x - 82.5)] + 2$$

4) The relationship between the stress on the shaft of an electric motor and time can be modelled with a sinusoidal function. Determine an equation of a function that describes stress in terms of time.



$$a = \frac{\text{max} - \text{min}}{2} = \frac{11 - 5}{2} = 3$$

$$k = \frac{360}{\text{period}} = \frac{360}{0.04} = 9000$$

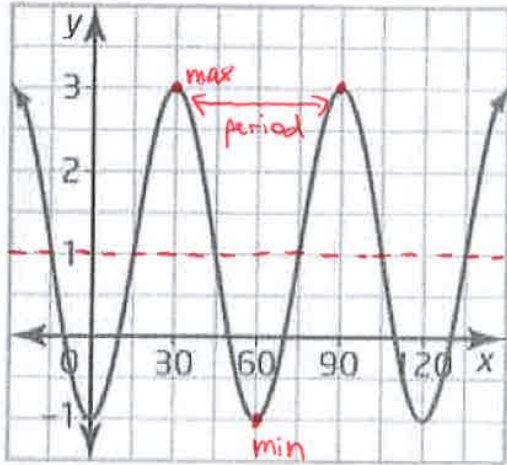
$$c = \text{max} - |a| = 11 - 3 = 8$$

$$d_{\sin} = 0$$

$$d_{\cos} = 0.01$$

$$y = 3 \sin[9000x] + 8 \quad \text{OR} \quad y = 3 \cos[9000(x - 0.01)] + 8$$

5) Determine the equation of the sine function shown.



$$a = \frac{\text{max} - \text{min}}{2} = \frac{3 - (-1)}{2} = 2$$

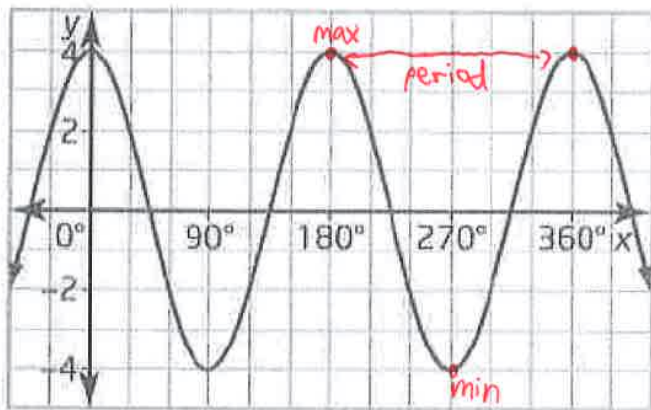
$$k = \frac{360}{\text{period}} = \frac{360}{60} = 6$$

$$c = \text{max} - |a| = 3 - 2 = 1$$

$$d_{\sin} = 15$$

$$y = 2 \sin[6(x - 15)] + 1$$

6) Represent the graph of the following functions using a sine and cosine function.



$$a = \frac{\text{max} - \text{min}}{2} = \frac{4 - (-4)}{2} = 4$$

$$k = \frac{360}{\text{period}} = \frac{360}{180} = 2$$

$$c = \text{max} - |a| = 4 - 4 = 0$$

$$d_{\cos} = 0$$

$$d_{\sin} = d_{\cos} - \frac{90}{k} = 0 - \frac{90}{2} = -45$$

$$y = 4 \cos(2x)$$

$$y = 4 \sin[2(x + 45)]$$

Answers

1) a) $y = 5 \sin[3(x + 30^\circ)] - 2$ b) $y = 5 \cos 3x - 2$

2) a) $y = \frac{1}{2} \sin\left[\frac{1}{2}(x + 180^\circ)\right] + 1$ b) $y = \frac{1}{2} \cos \frac{1}{2}x + 1$

3) $y = \cos[4(x - 82.5^\circ)] + 2$

4) $y = 3 \sin(9000x) + 8$ OR $y = 3 \cos[9000(x - 0.01)] + 8$

5) a) $y = 2 \sin[6(x - 15^\circ)] + 1$

b) If the maximum values are half as far apart, the period of the function is reduced by one-half to 30° . The value of k doubles from 6 to 12. The equation for the new function is $y = 2 \sin[12(x - 15^\circ)] + 1$.

6) $y = 4 \cos 2x$ and $y = 4 \sin[2(x + 45^\circ)]$.